Performance Optimization of FAS-Assisted U2G Communications Using Reinforcement Learning

Jianzhou Li, Zhentian Zhang, Wangqi Shi, and Hao Jiang, *Senior Member, IEEE*

***Abstract*—Fluid antenna systems (FAS) are emerging as a promising technology for sixth-generation (6G) networks due to their flexible and reconfigurable radiation characteristics. In this paper, we investigate the performance optimization of FAS-assisted unmanned aerial vehicle (UAV)-to-ground (U2G) communications. Unlike conventional static antenna configurations, the proposed framework leverages the unique ability of FAS to dynamically activate and adjust antenna ports, enabling optimization in U2G channels. A reinforcement learning approach based on an enhanced proximal policy optimization (PPO) algorithm, termed Memetic PPO, is introduced to efficiently explore the discrete FAS configuration space by integrating a local search mechanism. Key performance metrics, including channel capacity and modeling accuracy, are derived and analyzed. Simulation results demonstrate that the number of active ports and port spacing significantly affect both metrics, and the proposed Memetic PPO algorithm achieves superior configurations by effectively balancing these factors. These findings provide valuable insights for the design and optimization of FAS in practical UAV communication systems.**

***Index Terms*—Fluid antenna system, unmanned aerial vehicle, channel capacity, reinforcement learning, proximal policy optimization.**

I. Introduction

Despite the remarkable achievements of fifth-generation (5G), it still faces some shortcomings, such as an over-reliance on base stations and high costs. In contrast, sixth-generation (6G), as the next-generation communication technology, aims to integrate satellites to build a globally covered network and features low costs, holding promising development prospects. The fluid antenna system (FAS) is regarded as one of the key technologies for 6G and has demonstrated vigorous development momentum. The core concept of FAS lies in the reconfigurability of antenna shape and position, with similar ideas traceable back to the directionally reconfigurable waveguide antenna proposed in 1960. It was not until recent years that the concept of FAS was proposed in [1] and [2]. Essentially, FAS refers to any controllable fluid, conductive, or dielectric structure that can dynamically alter its shape and position to reconfigure key parameters such as gain, polarization, radiation pattern, and operating frequency, thereby enhancing signal transmission efficiency, spectrum utilization, and coverage. Currently, there exists substantial academic research on FAS. [3] indicated that under identical conditions, FAS exhibits higher secrecy capacity, lower outage probability, and superior energy efficiency compared to conventional antennas. Furthermore, several effective hardware implementations have emerged. For instance, the PB-FAS proposed in [4] offers low power consumption, low cost, and easy integration, making it suitable for large-scale smart city deployment. [5] found that in communication scenarios where base stations are equipped with fluid antennas, channel capacity can be enhanced by optimizing the antenna positions at both the base station and the user ends. [6] pointed out that when users are equipped with FAS, the scalability of the network can be improved by alleviating the base station's channel state information burden. [7] comprehensively presented the design, channel modeling, and experiments of an ER-FAS, the results demonstrated that their proposed ER-FAS performs effectively in both the near-field and far-field, thereby enhancing the overall spectral efficiency of the wireless communication system. Recent years have witnessed the rapid development of artificial intelligence (AI), and the deep integration of communication and AI aligns with the vision of 6G. [8] first introduced Bayesian regression into FAS and proposed S-BAR, a general, high-precision, low-pilot-overhead channel estimation method for FAS with strong robustness and adaptability.

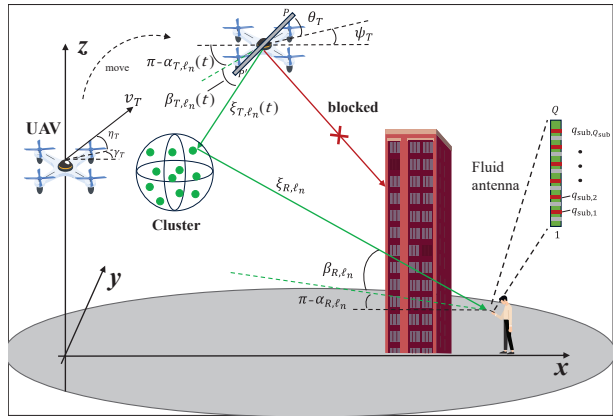

Fig. 1: Channel model of UAV-to-ground communication scenario assisted by FAS.

Although research on FAS is extensive, applying reinforcement learning (RL) to dynamic unmanned aerial vehicle (UAV)-to-ground (U2G) scenarios remains limited [8], [9]. To address the challenges of time-varying channels and the low exploration efficiency of baseline proximal policy optimization (PPO) in discrete action spaces, we propose an enhanced variant termed *Memetic PPO*. This algorithm

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J. Li and H. Jiang are with the College of Artificial Intelligence, Nanjing University of Information Science and Technology, 210044, P. R. China, H. Jiang is also with the School of Information Engineering, Jiangxi Provincial Key Laboratory of Advanced Signal Processing and Intelligent Communications, Nanchang University, Nanchang 330031, China (e-mails: 202412491494@nuist.edu.cn, jianghao@nuist.edu.cn).

Z. Zhang and W. Shi are with the National Mobile Communications Research Laboratory, Southeast University, Nanjing 210096, China (e-mails: zhentianzhangzzt@gmail.com, shiwangqi@seu.edu.cn.)

integrates a local search mechanism to refine action selection before execution, significantly accelerating convergence and uncovering superior configurations. Furthermore, considering the model error discussed in [10], our framework optimizes FAS configurations by balancing the trade-off between channel capacity and modeling accuracy, providing better solutions for practical UAV communication systems.

Notation: The lowercase, boldface lowercase, and boldface uppercase letters such as x, $\mathbf{x}$, and $\mathbf{X}$ stand for the scalars, vectors, and matries, respectively. $\|\cdot\|$ represents the Frobenius norm, $\|\cdot\|_\infty$ represents the ℓ_∞ norm and $(\cdot)^{\mathrm{H}}$ denotes the conjugate transpose operation.

II. System Model

Consider a three-dimensional (3D) channel model for A2G communications as shown in Fig. 1, the UAV moves at a speed of v_T, and the azimuth and elevation angles of the direction of the motion are γ_T and η_T, respectively. In our model, the projection of midpoint of the UAV on the ground is defined as the coordinate origin, and the direction pointing to the receiver on the ground is the positive direction of the x-axis; the z-axis goes straight up and y-axis follows the right-hand rule. It is worth mentioning that the coordinate system does not move along with the UAV. Define H_0 and D_0 as the initial altitude of the UAV and the distance from the origin to the receiver, respectively.

On the UAV end, the azimuth and elevation orientation angles of the P omni-directional uniform linear array (ULA) antennas equipped on the UAV are respectively described as ψ_T and θ_T. The position of the p-th element in ULA can be expressed as

$$\mathbf{d}_p(t) = \mathbf{d}_T(t) + \mathbf{d}_{T,p}, \tag{1}$$

where $\mathbf{d}_T(t)$ denotes the time-varying position of the UAV, and $\mathbf{d}_{T,p}$ denotes the distance vector from the ULA midpoint to the p-th antenna. To be specific, they can respectively be expressed as

$$\mathbf{d}_T(t) = v_T t \begin{bmatrix} \cos\eta_T \cos\gamma_T \\ \cos\eta_T \sin\gamma_T \\ \sin\eta_T \end{bmatrix} + \mathbf{d}_{t_0}, \tag{2}$$

$$\mathbf{d}_{T,p} = k_p \delta_T \begin{bmatrix} \cos\theta_T \cos\psi_T \\ \cos\theta_T \sin\psi_T \\ \sin\theta_T \end{bmatrix}, \tag{3}$$

where $k_p = (P - 2p + 1)/2$, $\mathbf{d}_{t_0} = [0, 0, H_0]^{\mathrm{T}}$ and δ_T is the distance between adjacent antennas.

The receiver on the ground is equipped with a fluid antenna with a length of $W\lambda$ and Q ports, where λ is the carrier wavelength and $Q \gg Q_{\text{sub}}$, with Q_{sub} being the number of activated ports. The adjacent ports spacing of the fluid antenna Δ can be expressed as

$$\Delta = \frac{W\lambda}{Q-1}. \tag{4}$$

To characterize the dynamic port activation, we define a mapping relationship between the activated port index and the physical port index. Specifically, the q_{sub}-th activated port ($q_{\text{sub}} = 1, 2, \ldots, Q_{\text{sub}}$) corresponds to the q-th physical port ($q = 1, 2, \ldots, Q$) in the FAS through a mapping function $q = \mathcal{M}(q_{\text{sub}})$. Therefore, the position of the q_{sub}-th activated port relative to the global coordinate origin can be expressed as

$$\mathbf{d}_q = \mathbf{d}_R + k_q \Delta \begin{bmatrix} \cos\theta_R \cos\psi_R \\ \cos\theta_R \sin\psi_R \\ \sin\theta_R \end{bmatrix}, \tag{5}$$

where $k_q = (Q - 2\mathcal{M}(q_{\text{sub}}) + 1)/2$, ψ_R, θ_R are respectively the azimuth and elevation orientation angles of the FAS, and $\mathbf{d}_R = [D_0, 0, 0]^{\mathrm{T}}$ accounts for the position of receiver relative to the global coordinate origin.

The signal transmitted from the UAV will go through non-line of sight (NLoS) components, where the signal undergoes scattering by the cluster before arriving at the receiver. Due to the mobility of the UAV, the departure angles from the scatterers become time-varying. For a fixed scatterer position $\mathbf{d}_{\ell n} = [x_{\ell n}, y_{\ell n}, z_{\ell n}]^{\mathrm{T}}$ in the global coordinate system, the relative position vector from the UAV to the scatterer is given by

$$\mathbf{d}_{T,\ell n}(t) = \mathbf{d}_{\ell n} - \mathbf{d}_T(t), \tag{6}$$

the time-varying azimuth and elevation angles of departure (AoD) are then computed as

$$\alpha_{T,\ell n}(t) = \arctan2\left(y_{\ell n} - [\mathbf{d}_T(t)]_y,\ x_{\ell n} - [\mathbf{d}_T(t)]_x\right), \tag{7}$$

$$\beta_{T,\ell n}(t) = \arcsin\left(\frac{z_{\ell n} - [\mathbf{d}_T(t)]_z}{\xi_{T,\ell n}(t)}\right), \tag{8}$$

where $\arctan2(\cdot,\cdot)$ denotes the four-quadrant inverse tangent function, and $\xi_{T,\ell n}(t) = \|\mathbf{d}_{T,\ell n}(t)\|$ is the Euclidean distance between the UAV and the scatterer, given by

$$\xi_{T,\ell n}(t) = \sqrt{\sum_{i\in\{x,y,z\}} \left([\mathbf{d}_{\ell n}]_i - [\mathbf{d}_T(t)]_i\right)^2}, \tag{9}$$

here, $[\cdot]_i$ denotes the i-th component of the vector in the global coordinate system.

We can represent the channel matrix for the proposed channel model as $\mathbf{H}^{\text{cluster}}(t,\tau)$, where τ denotes the path delay. Each element $h_{p,q_{\text{sub}}}(t,\tau)$ in $\mathbf{H}^{\text{cluster}}(t,\tau)$ stands for the complex channel impulse response (CIR) of the propagation link from the p-th element in ULA to the q_{sub}-th element in FAS, which can be given as

$$h_{p,q_{\text{sub}}}(t,\tau) = h^{\text{cluster}}_{p,q_{\text{sub}}}(t)\delta\left(\tau - \tau^{\text{cluster}}(t)\right), \tag{10}$$

where $\tau^{\text{cluster}}(t)$ is the path delay for the NLoS propagation link, and $h^{\text{cluster}}_{p,q_{\text{sub}}}(t)$ can be expressed as

$$\begin{aligned} h^{\text{cluster}}_{p,q_{\text{sub}}}(t) = \sum_{\ell\in L}\sum_{n=1}^{\ell_N} & e^{j\varphi_{\ell_n} - j\frac{2\pi}{\lambda}(\xi_{T,\ell_n}(t)+\xi_{R,\ell_n})} \\ & \times e^{j\frac{2\pi}{\lambda}k_p\delta_T\cos(\alpha_{T,\ell_n}(t)-\psi_T)\cos\beta_{T,\ell_n}(t)\cos\theta_T} \\ & \times e^{j\frac{2\pi}{\lambda}k_p\delta_T\sin\beta_{T,\ell_n}(t)\sin\theta_T} \\ & \times e^{j\frac{2\pi}{\lambda}k_q\Delta\cos(\alpha_{R,\ell_n}-\psi_R)\cos\beta_{R,\ell_n}\cos\theta_R} \\ & \times e^{j\frac{2\pi}{\lambda}k_q\Delta\sin\beta_{R,\ell_n}\sin\theta_R} \\ & \times e^{j\frac{2\pi}{\lambda}v_T t\cos(\alpha_{T,\ell_n}(t)-\gamma_T)\cos\beta_{T,\ell_n}(t)\cos\eta_T} \\ & \times e^{j\frac{2\pi}{\lambda}v_T t\sin\beta_{T,\ell_n}(t)\sin\eta_T}, \end{aligned} \tag{11}$$

where $\{\varphi_{\ell n}\}_{n=1,2,\ldots,N}$ represents the random phase, assumed to be independent and uniformly distributed as $\varphi_{\ell n} \sim \mathrm{U}[-\pi,\pi)$. The terms $\xi_{R,\ell n} = \|\mathbf{d}_{\ell_n} - \mathbf{d}_R\|$ denote distances between the receiver and the n-th path in the ℓ-th cluster. Furthermore, α_{R,ℓ_n} and β_{R,ℓ_n} are angles of arrival. The scatterer positions follow the von Mises distribution

$$f(\alpha) = \frac{e^{\kappa\cos(\alpha-\mu_\alpha)}}{2\pi I_0(\kappa)}, \tag{12}$$

where κ is the concentration parameter, μ_α is the mean value of α, and $I_0(\cdot)$ is the modified Bessel function of order 0.

For U2G wireless communication systems, channel capacity is a crucial indicator of the achievable spectral efficiency, which is given by

$$C = \log_2\left(\det\left(\mathbf{I}_{Q_{\text{sub}}} + \frac{\rho_{\text{SNR}}}{P}\overline{\mathbf{H}^{\text{cluster}}}(t,\tau)\overline{\mathbf{H}^{\text{cluster}}}^{\text{H}}(t,\tau)\right)\right), \tag{13}$$

where $\mathbf{I}_{Q_{\text{sub}}}$ denotes the identity matrix of size $Q_{\text{sub}} \times Q_{\text{sub}}$, ρ_{SNR} is the signal-to-noise ratio (SNR), and $\overline{\mathbf{H}^{\text{cluster}}}(t,\tau)$ represents the normalized channel matrix.

III. Memetic PPO Model

To solve the FAS configuration problem in U2G communications, we adopt the PPO algorithm as the core reinforcement learning framework. PPO is a policy-gradient method that achieves stable and efficient updates through clipped surrogate objectives. In our framework, the FAS optimization task can be formulated as a markov decision process (MDP) defined by the tuple $(\mathcal{S},\mathcal{A},\mathcal{P},R,\gamma)$, where $\mathcal{S}$ is the state space, $\mathcal{A}$ is the action space, $\mathcal{P}$ is the state transition probability, R is the reward function, and γ is the discount factor. The design of the state space $\mathcal{S}$ is crucial for bridging the physical channel model with the learning agent. Although the U2G channel exhibits complex characteristics as elaborated in Section II, its impact on receiver performance ultimately manifests in the statistical properties of the effective channel matrix $\mathbf{H}_{\text{FAS}}^{\text{cluster}}(t,\tau)$. Specifically, the rank and condition number of this matrixłwhich directly determine the achievable capacityłare governed by the antenna length W, the number of activated ports Q_{sub}, and the resulting port spacing Δd. Therefore, we select these three directly configurable physical quantities as the state representation, expressed as

$$\mathbf{s} = [W, Q_{\text{sub}}, \Delta d] \in \mathcal{S}. \tag{14}$$

This representation enables the agent to indirectly influence both channel capacity and modeling error by dynamically adjusting the FAS hardware configuration. The action space $\mathcal{A}$ is a multi-dimensional discrete space comprising two independent components: the FAS length adjustment action a_W for modifying the normalized physical length W, and the activated ports adjustment action $a_{Q_{\text{sub}}}$ for changing the port count. The complete action vector is defined as

$$\mathbf{a} = [a_W, a_{Q_{\text{sub}}}] \in \mathcal{A}. \tag{15}$$

The reward function is designed to balance multiple objectives and takes the following form

$$R(C,\delta_{\text{error}}) = r_{\text{c}}C + r_{\text{error}}\delta_{\text{error}}, \tag{16}$$

where C is the instantaneous channel capacity, δ_{error} is the modeling error between FAS and ULA configurations, and r_{c} and r_{error} are their respective weighting coefficients. At each time step t, the agent observes the current state $\mathbf{s}_t \in \mathcal{S}$ and selects an action $\mathbf{a}_t \in \mathcal{A}$ according to the stochastic policy $\pi_\theta(\mathbf{a}_t|\mathbf{s}_t)$ parameterized by θ. The environment then executes the action, updates the FAS configuration, computes the channel matrix, and returns the next state $\mathbf{s}_{t+1}$ along with the reward r_t defined in (17). The objective of PPO is to maximize the expected discounted return

$$J(\theta) = \mathbb{E}_{\pi_\theta}\left[\sum_{t=0}^{T}\gamma^t r_t\right]. \tag{17}$$

To ensure stable policy updates, PPO introduces a clipped surrogate objective. Let

$$r_t(\theta) = \frac{\pi_\theta(\mathbf{a}_t|\mathbf{s}_t)}{\pi_{\theta_{\text{old}}}(\mathbf{a}_t|\mathbf{s}_t)} \tag{18}$$

denote the probability ratio between the updated and old policies. The clipped objective is defined as

$$\mathcal{L}^{\text{PPO}}(\theta) = \mathbb{E}_t\left[\min\left(r_t(\theta)\hat{A}_t, \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t\right)\right], \tag{19}$$

where $\hat{A}_t$ is the advantage estimate obtained via Generalized Advantage Estimation (GAE), and ϵ is the clipping parameter. This mechanism prevents excessively large policy updates, thereby enhancing convergence stability and avoiding destructive policy oscillations.

The PPO-based FAS optimization procedure begins with initializing the policy and value networks. The agent gathers interaction trajectories by following the current policy, computes rewards based on channel capacity and modeling error, and estimates advantages using generalized advantage estimation. Policy parameters are updated by maximizing the clipped surrogate objective, while the value function is refined by minimizing the value loss. To overcome the limitations of stochastic trial-and-error exploration in baseline PPO, we propose an improved variant called *memetic PPO*. This method leverages the spatial correlation inherent in FAS port configurations, specifically, the similarity of channel characteristics among adjacent ports. By incorporating such physical priors into action selection, memetic PPO evolves from blind exploration to a physics-aware refinement strategy. For a candidate action $\mathbf{a}_t$ generated by the policy network, a local search is conducted within a predefined discrete neighborhood of radius r:

$$\mathcal{N}(\mathbf{a}_t) = \{\mathbf{a}' \mid \|\mathbf{a}' - \mathbf{a}_t\|_\infty \le r\}, \tag{20}$$

where $\mathbf{a}'$ denotes a candidate action within the neighborhood of $\mathbf{a}_t$. All candidate actions within this neighborhood are evaluated using the immediate reward function while keeping the environment state unchanged. The refined action is then selected as

$$\mathbf{a}_t^* = \arg\max_{\mathbf{a}'\in\mathcal{N}(\mathbf{a}_t)} R(\mathbf{s}_t, \mathbf{a}'), \tag{21}$$

where $\mathbf{a}_t^*$ represents the refined actionłthe locally optimal choice within $\mathcal{N}(\mathbf{a}_t)$ that maximizes the immediate reward under the current state $\mathbf{s}_t$.

The key distinction between baseline PPO and Memetic PPO is that the former relies solely on gradient-based stochastic exploration, whereas Memetic PPO integrates policy gradients with deterministic neighborhood search. This hybrid structure introduces evolutionary-inspired local refinement into the reinforcement learning loop, yielding several advantages. First, it significantly accelerates convergence, as the neighborhood search allows the algorithm to quickly locate locally optimal configurations within the discrete FAS parameter space. Second, in discrete problems such as FAS configuration adjustment, neighboring configurations often exhibit strong performance correlation; local search effectively exploits this structural property to achieve finer-grained optimization than random sampling. Finally, it improves training stability by reducing reward variance through local refinement, leading to smoother learning curves and more stable convergence behavior.

IV. Results and Analysis

The experiment employs realistic parameters for UAV-to-ground communications: carrier frequency $f_c = 5$ GHz, UAV initial altitude $H_0 = 20$ m, initial horizontal distance $D_0 = 50$ m, number of UAV antennas $P = 4$, total FAS ports $Q = 30$, UAV antenna spacing $\delta_T = \lambda/2$, UAV azimuth orientation angle $\psi_T = \pi/2$, UAV elevation orientation angle $\theta_T = \pi/3$, UAV velocity $v_T = 3$ m/s, UAV motion azimuth angle $\gamma_T = \pi/3$, UAV motion elevation angle $\eta_T = \pi/3$, and fixed time instant $t = 2$ s. The scattering environment consists of $L = 3$ clusters with 10 paths per cluster, following a von Mises distribution with an environment factor $\kappa = 2.0$. Training utilizes the PPO algorithm with the following hyperparameters: learning rate 1×10^{-4}, steps per update 1024, batch size 128, epochs per update 4, discount factor $\gamma = 0.99$, generalized advantage estimation parameter $\lambda = 0.95$, clipping parameter $\epsilon = 0.2$, entropy coefficient 0.03, value function coefficient 0.5, and maximum gradient norm 1.0. The policy network comprises two hidden layers with 128 neurons each, using ReLU activation functions. For the memetic PPO, a local search radius of 1 is employed to refine actions before execution. Training proceeds for 20,480 timesteps, with evaluation performed every 2,048 timesteps using 16 episodes. The environment's maximum episode length is set to 64 steps to ensure efficient exploration. Model checkpoints are saved based on evaluation performance, and training progress is monitored through reward progression and modeling error metrics.

By setting $r_{\text{error}} = 0$ in (17), which corresponds to the scenario where the algorithm optimizes exclusively for channel capacity, the baseline PPO converges to an optimal FAS configuration of $W = 15.0$ and $Q_{\text{sub}} = 30$. In contrast, the memetic PPO yields a slightly different configuration of $W = 16.0$ and $Q_{\text{sub}} = 30$. The relationship between channel capacity and the parameters W and Q_{sub} is illustrated in Fig. 2. The simulation results reveal that for a fixed antenna length W, the channel capacity exhibits a significant upward trend as Q_{sub} increases. This phenomenon is primarily attributed to the fact that activating additional ports enhances the rank

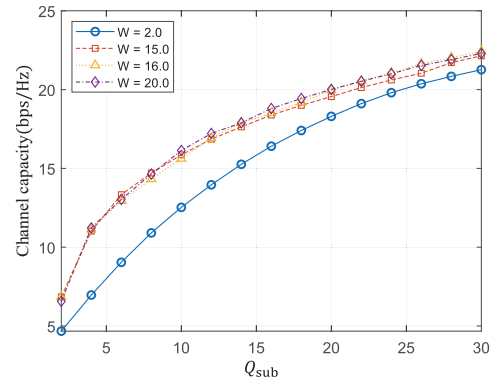

Fig. 2: Channel capacity of different FAS configurations

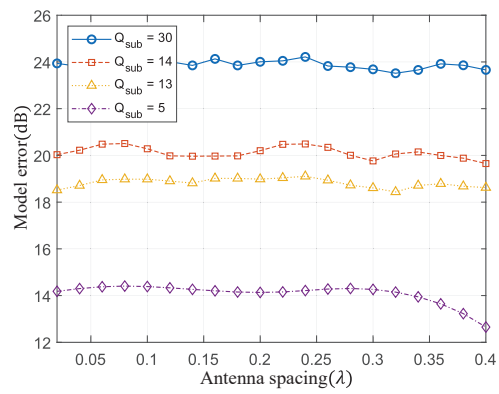

Fig. 3: Model errors with different FAS configurations

of the channel matrix $\mathbf{H}_{\text{FAS}}^{\text{cluster}}(t, \tau)$, which, according to (13), serves as a fundamental determinant of capacity. Specifically, the expanded spatial degrees of freedom facilitate richer multipath exploitation, thereby improving spectral efficiency. This observation is consistent with the findings in [10] and [11], corroborating the validity of our channel modeling and optimization results. Notably, while both algorithms identify the maximum number of ports ($Q_{\text{sub}} = 30$) as optimal, the memetic PPO attains a superior capacity by refining the antenna length to $W = 16.0$. This performance gain stems from the local search mechanism inherent in memetic PPO, which effectively explores the neighborhood of candidate actions to escape suboptimal solutions.

Correspondingly, by configuring $r_c = 0$ in (17), the optimization focuses exclusively on minimizing modeling error. Under this configuration, the baseline PPO identifies an optimal FAS setup of $W = 11.0$ and $Q_{\text{sub}} = 14$, while memetic PPO refines this to $Q_{\text{sub}} = 13$. Fig. 3 illustrates the modeling error as a function of Q_{sub} and Δd for $W = 11.0$. Consistent with (14), the error increases monotonically with Q_{sub}. This occurs because denser port activation within a fixed spatial aperture exacerbates the deviation from the benchmark ULA channel characteristics. Notably, both variants converge to configurations with Q_{sub} values that are higher than the $Q_{\text{sub}} = 5$ configuration, which was observed in simulations to yield lower modeling error. This sub-optimality stems from two factors: (i) the non-stationary channel induced by UAV mobility creates a "narrow basin of attraction" for the

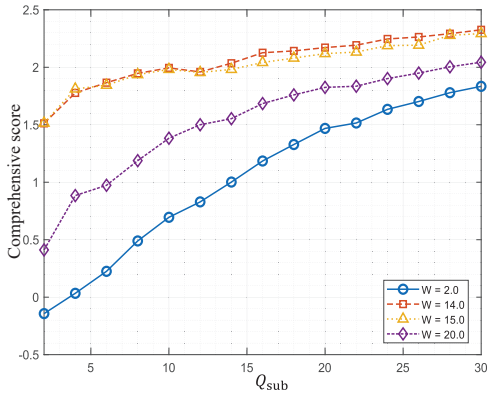

Fig. 4: Comprehensive score with different FAS configurations

rewards at extremely low Q_{sub}, making these optima difficult to sustain under stochastic exploration; and (ii) the PPO clipping mechanism, while ensuring stability, may bias the agent toward broader, more robust local optima that require less search pressure. Nevertheless, memetic PPO achieves a lower error than the baseline, demonstrating that its integrated local search effectively enhances exploration within the discrete action space to locate superior configurations.

To evaluate the system performance under a joint optimization framework, the weighting factors in (17) are empirically assigned as $r_c = 0.2$ and $r_{\text{error}} = -0.1$. These coefficients are normalized to enhance numerical stability during the training process, with the relative magnitudes reflecting a prioritized focus on maximizing channel capacity while simultaneously penalizing modeling inaccuracies. Under this multi-objective setting, the baseline PPO converges to a configuration of $W = 15.0$ and $Q_{\text{sub}} = 30$, whereas the memetic PPO identifies a more refined solution of $W = 14.0$ and $Q_{\text{sub}} = 30$. The functional relationship between the comprehensive score and the FAS parameters is illustrated in Fig. 4. The simulation results demonstrate that the composite utility increases monotonically with the number of activated ports. Although an increase in Q_{sub} inevitably exacerbates the modeling error [10], the chosen weighting strategy ensures that the gains in spectral efficiency dominate the overall performance metric. Specifically, the positive reward for capacity outweighs the penalty incurred by modeling deviations, thereby favoring configurations with high Q_{sub}. Furthermore, the configuration attained by the memetic PPO ($W = 14.0$) yields a superior comprehensive score compared to the baseline PPO's result ($W = 15.0$). This performance margin is attributable to the hybrid optimization architecture of memetic PPO, which synergistically combines stochastic gradient-based policy updates with deterministic local search to navigate the complex multi-objective landscape more effectively. The discrepancy in the identified optimal antenna length suggests that memetic PPO is capable of discovering a more balanced equilibriumłwhere a moderate antenna length maintains high capacity while effectively managing system complexity. This underscores the algorithm's robustness in capturing nuanced trade-offs that remain elusive to the standard PPO framework.

V. Conclusion

This paper has investigated the performance optimization problem for a FAS-assisted U2G communication system using reinforcement learning. The proposed framework can adaptively configure FAS parameters, such as antenna length and the number of active ports, in dynamic UAV scenarios, exploring the potential of data-driven methods for enhancing U2G communication performance. Through comprehensive simulation and comparative analysis, the effectiveness of the PPO algorithm and its memetic variant has been validated. It has been demonstrated that the number of active ports and the fluid antenna length significantly impact the trade-off between channel capacity and modeling error. Specifically, while increasing the number of active ports leads to higher channel capacity, it simultaneously elevates the modeling error, emphasizing the requirement for an intelligent configuration strategy. Moreover, the memetic PPO variant is shown to achieve superior comprehensive scores and better convergence stability compared to the standard PPO by integrating deterministic local search. Finally, it should be pointed out that applying this RL-based framework to multi-user interference scenarios and incorporating practical constraints like energy efficiency are key research directions in our future research.

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